

- FEDERICO MALLEA* AND MARCELO CONIGLIO, *On the decidability of self-extensional logics of formal inconsistency.*

Universidade Estadual de Campinas (UNICAMP), Centro de Lógica, Epistemologia e História da Ciência (CLE).

E-mail: hfmallea@gmail.com.

RmbC is a logic belonging to the family of paraconsistent logics called *Logics of Formal Inconsistency* (LFIs). It is characterized by a non-explosive negation $\neg$ and a consistency operator $\circ$ that allows to recover the explosion principle in a controlled manner. RmbC is an extension of mbC, the basic LFI, obtained by the addition of the replacement property by means of two (global) inference rules. This logic, along with a family of self-extensional LFIs that are axiomatic extensions of RmbC, was introduced in 2021 by Carnielli, Coniglio, and Fuenmayor.

The decidability of RmbC and these axiomatic extensions remains an open question. This talk presents progress on this important open problem by showing that RmbC is decidable by means of BALFIs, its natural class of algebraic models. BALFIs (Boolean algebras with LFI operators) are expansions of Boolean algebras with two unary operators interpreting $\neg$ and $\circ$. The decidability of RmbC was obtained via a constructive proof of the finite model property (FMP) for RmbC, adapted from ideas in Harrop's classic 1958 paper. The construction is effective and allows the study of validity in RmbC to be reduced to the evaluation of a finite family of finite models.

In addition, a natural limitation of the method for some axiomatic extensions of RmbC is observed. Indeed, in some systems (notably, when $\neg\neg A \leftrightarrow A$ holds) the paraconsistent negation $\neg$ necessarily collapses to the Boolean complement in finite models, preventing the construction of finite paraconsistent BALFI models. This means that paraconsistent axiomatic extensions of RmbC validating this axiom do not have the FMP w.r.t. the variety of BALFIs. Hence, decidability of such logics (if valid) must be proven either by proof-theoretical methods or by using another class of models.

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Keywords: LFIs, LFIs with replacement, paraconsistency, decidability, finite model property, BALFIs, self-extensionality.

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